

Numerical Algorithms — Exam Cheatsheet

Duality · Convexity · Numerical Differentiation · Integration · Optimization · Sparse Matrices · RL

All formulas, theorems, and proofs you need.

-21. NUMERICAL DIFFERENTIATION

Finite Difference Formulas

Forward difference — error $O(h)$:

$$D_f(h) = \frac{f(x+h) - f(x)}{h} = f'(x) + \frac{h}{2}f''(x) + O(h^2)$$

Backward difference — error $O(h)$:

$$D_b(h) = \frac{f(x) - f(x-h)}{h} = f'(x) - \frac{h}{2}f''(x) + O(h^2)$$

Central difference — error $O(h^2)$ (much better!):

$$D_c(h) = \frac{f(x+h) - f(x-h)}{2h} = f'(x) + \frac{h^2}{6}f'''(x) + O(h^4)$$

Second derivative — error $O(h^2)$:

$$f''(x) \approx \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}$$

4th-order formula D_4 — error $O(h^4)$:

$$D_4(h) = \frac{-f(x+2h) + 8f(x+h) - 8f(x-h) + f(x-2h)}{12h}$$

Richardson Extrapolation

Used to cancel the leading error term and boost accuracy.

If $D(h) = f'(x) + Ch^2 + O(h^4)$ (e.g. central diff), then:

$$D_{\text{ext}} = \frac{4D(h/2) - D(h)}{3} \Rightarrow \text{error } O(h^4)$$

General rule: If error $= Ch^p$, eliminate it via:

$$\frac{2^p D(h/2) - D(h)}{2^p - 1}$$

Competing Errors & Optimal Step Size

Two errors trade off as h changes:

- **Truncation error:** decreases as $h \rightarrow 0$ (formula approximation)
- **Rounding/cancellation error:** increases as $h \rightarrow 0$ (finite precision)

Formula	Optimal h^*	Min error
Forward diff	$\sqrt{2\varepsilon_m}$	$\sim \sqrt{2\varepsilon_m} f'' $
Central diff	$\varepsilon_m^{1/3}$	$\sim \varepsilon_m^{2/3}$

$\varepsilon_m \approx 10^{-16}$ (double precision machine epsilon).

Central diff is $\approx 10\times$ more accurate than forward for same h .

-22. NUMERICAL INTEGRATION

Composite Trapezoidal Rule — error $O(h^2)$

Divide $[a, b]$ into n equal panels of width $h = (b-a)/n$:

$$\int_a^b f dx \approx \frac{h}{2} \left[f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n) \right]$$

Global error: $-\frac{(b-a)h^2}{12}f''(c)$ for some $c \in [a, b]$.

Composite Simpson's Rule — error $O(h^4)$

Coefficients follow the pattern 1, 4, 2, 4, 2, ..., 4, 2, 4, 1:

$$\int_a^b f dx \approx \frac{h}{3} [f_0 + 4f_1 + 2f_2 + 4f_3 + \dots + 4f_{n-1} + f_n]$$

Requires n to be even.

Global error: $-\frac{(b-a)h^4}{180}f^{(4)}(c)$.

Exact for polynomials of degree ≤ 3 .

Gaussian Quadrature — highest accuracy per evaluation

Transform $[a, b]$ to $[-1, 1]$ via $x = \frac{b-a}{2}t + \frac{a+b}{2}$:

$$\int_a^b f(x) dx \approx \frac{b-a}{2} \sum_{i=1}^n w_i f\left(\frac{b-a}{2}t_i + \frac{a+b}{2}\right)$$

2-point Gauss-Legendre (exact for degree ≤ 3 , i.e. $2n-1=3$):

$$t_{1,2} = \pm \frac{1}{\sqrt{3}}, \quad w_1 = w_2 = 1$$

3-point Gauss-Legendre (exact for degree ≤ 5):

$$t_{1,3} = \pm \sqrt{\frac{3}{5}}, \quad t_2 = 0, \quad w_{1,3} = \frac{5}{9}, \quad w_2 = \frac{8}{9}$$

n nodes $\Rightarrow$ exact for polynomials of degree $\leq 2n-1$.

Romberg Integration

Apply Richardson extrapolation repeatedly to trapezoidal estimates:

$$R(n, m) = \frac{4^m R(n, m-1) - R(n-1, m-1)}{4^m - 1}$$

where $R(n, 0)$ = trapezoidal rule with 2^n panels. Diagonal $R(n, n)$ converges very fast.

Comparison Table

-23. STANDARD FORM & BASIC DEFINITIONS		
Method	Global error	Notes
Forward/Backward diff	$O(h)$	Cheapest
Central diff	$O(h^2)$	Preferred
Trapezoidal	$O(h^2)$	Simple
Simpson's	$O(h^4)$	Need n even
GL Quadrature (n pts)	exact deg $2n - 1$	Fewest fn ev
Romberg	$O(h^{2^k})$	Builds on tr

Standard Form of Optimization

$$\begin{aligned} \min_x \quad & f_0(x) \\ \text{subject to} \quad & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_j(x) = 0, \quad j = 1, \dots, p \end{aligned}$$

- **Optimization variable:** $x \in \mathbb{R}^n$
- **Domain:** $\mathcal{D} = \text{dom}(f_0) \cap \bigcap_i \text{dom}(f_i) \cap \bigcap_j \text{dom}(h_j)$
- **Feasible set:** $\mathcal{F} = \{x \in \mathcal{D} : f_i(x) \leq 0 \ \forall i, \ h_j(x) = 0 \ \forall j\}$
- **Optimal value:** $p^* = \inf_{x \in \mathcal{F}} f_0(x)$
- $p^* = +\infty$ if infeasible; $p^* = -\infty$ if unbounded below
- **Optimal point** x^* : feasible and $f_0(x^*) = p^*$
- **Attained:** the infimum is achieved (a minimizer exists)

Constraint Terminology

At a feasible point $\bar{x}$:

- **Active** (tight): $f_i(\bar{x}) = 0$ (constraint is binding)
- **Inactive:** $f_i(\bar{x}) < 0$ (constraint has slack)
- **Redundant:** removing it does not change $\mathcal{F}$
- ϵ -suboptimal: $\tilde{x} \in \mathcal{F}$ with $f_0(\tilde{x}) \leq p^* + \epsilon$

Standard Conversions

- **Max to min:** $\max f_0 \Leftrightarrow \min(-f_0)$
- **Flip inequality:** $g(x) \geq 0 \Rightarrow -g(x) \leq 0$
- **Box constraints:** $l_i \leq x_i \leq u_i \Rightarrow x_i - u_i \leq 0$ and $l_i - x_i \leq 0$
- **Slack variable:** $f_i(x) \leq 0 \Rightarrow f_i(x) + s_i = 0, \ s_i \geq 0$

Singularities

If $f(x) \rightarrow \infty$ as $x \rightarrow a$: use a substitution to smooth the integrand.
E.g., $x = a + t^2$ maps the boundary behavior to a finite limit.

| -24. CONVEX SETS | | |
| **Definition:** $C \subseteq \mathbb{R}^d$ is **convex** if for all $x, y \in C$ and $\theta \in [0, 1]$: $$\theta x + (1 - \theta)y \in C$$ The line segment between any two points stays inside C . **Examples:** halfspaces $\{x : a^\top x \leq b\}$, balls $\{x : \|x\| \leq r\}$, polyhedra, $\mathbb{R}^n$ itself. **Intersection property:** **Intersection of any collection of convex sets is convex.** (Proof: if x, y are in all sets, so is any convex combo — by definition of each set.) **Why it matters:** Feasible sets of convex programs are intersections of convex sets (each $\{x : f_i(x) \leq 0\}$ is convex when f_i is convex). So $\mathcal{F}$ is convex. | | |
| -25. CONVEX FUNCTIONS | | |
| **Three Equivalent Definitions** **(0th order / Definition):** $\text{dom}(f)$ is convex and: $$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y) \quad \forall x, y, \theta \in [0, 1]$$ The graph lies *below* every chord. | | |

(1st order): If f is differentiable, f is convex $\Leftrightarrow$

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) \quad \forall x, y \in \text{dom}(f)$$

Tangent plane is a global underestimator.

(2nd order): If f is twice differentiable, f is convex $\Leftrightarrow$

$$\nabla^2 f(x) \succeq 0 \quad \forall x \in \text{dom}(f)$$

(Hessian is positive semidefinite everywhere.)

Key Operations that Preserve Convexity

- **Pointwise supremum:** $F(x) = \sup_{y \in \mathcal{Y}} f(x; y)$ is convex if each $f(\cdot; y)$ is convex.

Proof: For each fixed y : $f(\theta x_1 + (1 - \theta)x_2; y) \leq \theta f(x_1; y) + (1 - \theta)f(x_2; y)$.

Take $\sup_y$ of both sides.

- **Affine composition:** If f is convex, then $g(x) = f(Ax + b)$ is convex.

Proof: $A(\theta x_1 + (1 - \theta)x_2) + b = \theta(Ax_1 + b) + (1 - \theta)(Ax_2 + b)$.

Apply convexity of f .

- **Non-negative sum:** $\alpha_1 f_1 + \alpha_2 f_2$ is convex if f_i convex, $\alpha_i \geq 0$.

Common Convex / Concave Functions

Function	Type
$x^2, e^x, x , \ x\ _p$ ($p \geq 1$)	Convex
x^p for $p \geq 1$ or $p \leq 0$ ($x > 0$)	Convex
$x^\top Ax$ with $A \succeq 0$	Convex
$\frac{1}{2}\ Ax - b\ _2^2$ (Hessian = $A^\top A \succeq 0$)	Convex
$\log x, \sqrt{x}, -x \log x$	Concave
Affine $ax + b$	Both convex and concave

Indicator Function

$$\mathbf{1}_X(x) = \begin{cases} 0 & \text{if } x \in X \\ +\infty & \text{if } x \notin X \end{cases}$$

$\mathbf{1}_X$ is convex $\Leftrightarrow X$ is convex.

Proof ($\Rightarrow$): Take $x, y \in X$. Then $\mathbf{1}_X(x) = \mathbf{1}_X(y) = 0$. By convexity of $\mathbf{1}_X$: $\mathbf{1}_X(\theta x + (1 - \theta)y) \leq \theta \cdot 0 + (1 - \theta) \cdot 0 = 0$. Since $\mathbf{1}_X \geq 0$, this forces the value to be 0, so the combo is in X .

Proof ($\Leftarrow$): Case 1: $x \notin X$ or $y \notin X$ — RHS = $+\infty$, holds trivially. Case 2: $x, y \in X$ — since X is convex, $\text{combo} \in X$, so LHS = $0 = \text{RHS}$.

Reformulation:

$$\min_{x \in X} f(x) = \min_{x \in \mathbb{R}^n} [f(x) + \mathbf{1}_X(x)]$$

2. $\|\alpha x\| = |\alpha| \|x\|$ for all $\alpha \in \mathbb{R}$

3. $\|x + y\| \leq \|x\| + \|y\|$ (triangle inequality)

Dual norm:

$$\|y\|_* = \sup\{y^\top x : \|x\| \leq 1\} = \sup_{x \neq 0} \frac{y^\top x}{\|x\|}$$

Norm	Dual norm	Conjugate exponents
ℓ_2	ℓ_2 (self-dual)	$p = q = 2$
ℓ_1	ℓ_∞	$p = 1, q = \infty$
ℓ_∞	ℓ_1	$p = \infty, q = 1$
ℓ_p	ℓ_q	$\frac{1}{p} + \frac{1}{q} = 1$

Generalized Cauchy-Schwarz: $y^\top x \leq \|y\|_* \|x\|$

Hölder's inequality: $\sum_i |x_i y_i| \leq \|x\|_p \|y\|_q$ where $\frac{1}{p} + \frac{1}{q} = 1$

-26. NORMS & DUAL NORMS

Norm axioms: $\|\cdot\|$ is a norm if:

1. $\|x\| \geq 0$, and $\|x\| = 0 \Leftrightarrow x = 0$

-27. LAGRANGIAN & DUALITY

The Lagrangian

$$L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{j=1}^p \nu_j h_j(x)$$

Require $\lambda_i \geq 0$ for inequality constraints. No sign restriction on ν_j .

The Dual Function

$$g(\lambda, \nu) = \inf_{x \in \mathcal{D}} L(x, \lambda, \nu)$$

- g is always **concave** in (λ, ν) — even for nonconvex problems (it is the pointwise infimum of affine functions of (λ, ν)).
- $g(\lambda, \nu) \leq p^*$ for all $\lambda \geq 0$ — this is **weak duality**.

Weak Duality — Proof

For any $\lambda \geq 0$, any ν , and any feasible $\tilde{x}$ (so $f_i(\tilde{x}) \leq 0$ and $h_j(\tilde{x}) = 0$):

$$\begin{aligned} g(\lambda, \nu) &= \inf_x L(x, \lambda, \nu) \\ &\leq L(\tilde{x}, \lambda, \nu) \quad (\text{inf} \leq \text{any particular value}) \\ &= f_0(\tilde{x}) + \underbrace{\sum_i \lambda_i f_i(\tilde{x})}_{\leq 0 \text{ since } \lambda_i \geq 0, f_i \leq 0} + \underbrace{\sum_j \nu_j h_j(\tilde{x})}_{= 0} \\ &\leq f_0(\tilde{x}) \end{aligned}$$

Since this holds for every feasible $\tilde{x}$, take the infimum over feasible $\tilde{x}$:

$$\boxed{g(\lambda, \nu) \leq p^* \quad \text{for all } \lambda \geq 0, \nu} \quad \square$$

Dual Problem

$$\max_{\lambda \geq 0, \nu} g(\lambda, \nu)$$

Dual optimal value: $d^* = \max_{\lambda \geq 0, \nu} g(\lambda, \nu)$.

Duality gap: $p^* - d^* \geq 0$ always. Zero iff strong duality.

Strong Duality — Slater's Condition

Strong duality $d^* = p^*$ holds when:

- (1) Problem is **convex**: f_0, f_i convex, h_j affine.
- (2) **Slater's condition**: $\exists \tilde{x} \in \text{relint}(\mathcal{D})$ with $f_i(\tilde{x}) < 0$ for all i .

Concrete Example: $\min x^\top x$ subject to $Ax = b$

Form the Lagrangian (only equality constraints, so dual variable $\nu \in \mathbb{R}^p$):

$$L(x, \nu) = \|x\|_2^2 + \nu^\top (Ax - b)$$

Minimize over x (set gradient to zero):

$$\nabla_x L = 2x + A^\top \nu = 0 \Rightarrow x^* = -\frac{1}{2} A^\top \nu$$

Substitute x^* back:

$$\begin{aligned} g(\nu) &= \|x^*\|^2 + \nu^\top A x^* - \nu^\top b \\ &= \frac{1}{4} \nu^\top A A^\top \nu - \frac{1}{2} \nu^\top A A^\top \nu - \nu^\top b \end{aligned}$$

$$\boxed{g(\nu) = -\frac{1}{4} \nu^\top A A^\top \nu - \nu^\top b} \quad (\text{concave in } \nu)$$

Primal minimizer: $x^* = A^\top (A A^\top)^{-1} b$ (minimum-norm solution).

Primal optimal: $p^* = b^\top (A A^\top)^{-1} b$. Strong duality holds.

-28. KKT CONDITIONS

All four conditions must hold at a primal-dual optimal (x^*, λ^*, ν^*) :

1. Primal feasibility:

$$f_i(x^*) \leq 0 \quad \forall i, \quad h_j(x^*) = 0 \quad \forall j$$

2. Dual feasibility:

$$\lambda_i^* \geq 0 \quad \forall i$$

3. Complementary slackness:

$$\lambda_i^* \cdot f_i(x^*) = 0 \quad \forall i$$

Meaning: either the constraint is *active* ($f_i(x^*) = 0$), or its multiplier is zero ($\lambda_i^* = 0$). Both can also hold.

4. Stationarity (gradient of Lagrangian = 0):

$$\nabla f_0(x^*) + \sum_{i=1}^m \lambda_i^* \nabla f_i(x^*) + \sum_{j=1}^p \nu_j^* \nabla h_j(x^*) = 0$$

For convex problems with Slater's condition: KKT conditions are **necessary and sufficient** for global optimality. To find the solution: solve the 4 KKT equations simultaneously.

How to Use KKT in Practice

1. Write stationarity condition ($\nabla_x L = 0$).
2. Use complementary slackness: guess which constraints are active. Try $\lambda_i = 0$ for each inactive constraint.
3. Solve the resulting system.
4. Verify dual feasibility $\lambda_i \geq 0$ and primal feasibility.

-29. FENCHEL CONJUGATES

Definition

$$f^*(y) = \sup_{x \in \text{dom}(f)} [y^\top x - f(x)]$$

- f^* is always **convex** in y — regardless of whether f is convex (it's a sup of affine functions).
- **Fenchel inequality:** $f(x) + f^*(y) \geq x^\top y$ for all x, y .

- **Fenchel-Moreau theorem:** If f is proper, closed, and convex, then $f^{**} = f$.
- For **nonconvex** f : f^{**} equals the **convex envelope** of f (the largest convex function $\leq f$ everywhere).

How to Compute $f^*(y)$

1. Form $\phi(x) = y^\top x - f(x)$.
2. Find the maximizer: set $\frac{d\phi}{dx} = y - f'(x) = 0$, solve for $x^*(y)$.
3. Substitute: $f^*(y) = y^\top x^*(y) - f(x^*(y))$.

Worked Example: $f(x) = x^2$

Step 1 — compute f^* :

$$f^*(y) = \sup_x [xy - x^2]$$

Set $\frac{d}{dx}[xy - x^2] = y - 2x = 0 \Rightarrow x^* = \frac{y}{2}$.

Substitute: $f^*(y) = y \cdot \frac{y}{2} - \left(\frac{y}{2}\right)^2 = \frac{y^2}{2} - \frac{y^2}{4} = \frac{y^2}{4}$.

Step 2 — compute f^{} and verify:**

$$f^{**}(x) = \sup_y \left[xy - \frac{y^2}{4} \right]$$

Set $x - \frac{y}{2} = 0 \Rightarrow y^* = 2x$. Substitute:

$$f^{**}(x) = x(2x) - \frac{(2x)^2}{4} = 2x^2 - x^2 = x^2 = f(x) \quad \checkmark$$

Fenchel-Moreau confirmed: f is convex, so $f^{**} = f$.

Key Conjugate Pairs

$f(x)$	$f^*(y)$	Domain of f^*
x^2	$y^2/4$	$\mathbb{R}$
$\frac{1}{2}x^2$	$\frac{1}{2}y^2$	$\mathbb{R}$ (self-conjugate)
$ x $	$\mathbf{1}_{[-1,1]}(y)$	$ y \leq 1$, else $+\infty$
$-\log x$	$-1 - \log(-y)$	$y < 0$
e^x	$y \log y - y$	$y > 0$
$\ x\ _2$	$\mathbf{1}_{\ y\ _2 \leq 1}$	$\ y\ _2 \leq 1$, else $+\infty$
$ax + b$	$-b$ if $y = a$, else $+\infty$	$y = a$ only

Biconjugate and Convex Envelope

For any function f : $f^{**}(x) \leq f(x)$ everywhere.

For a **nonconvex** f (e.g. double-well $f(x) = (x^2 - 1)^2$): f^{**} is obtained by “filling in” the nonconvex regions — it equals the convex hull of the epigraph of f .

Practical use: Replacing f by f^{**} gives a convex relaxation with optimal value $\leq p^*$.

210. GRADIENT DESCENT & ALGORITHMS

Gradient Descent

$$x_{k+1} = x_k - \eta_k \nabla f(x_k)$$

- Direction $-\nabla f(x_k)$ is the steepest descent direction.
- **Fixed step size η :** converges if $\eta < \frac{2}{\lambda_{\max}(\nabla^2 f)}$.

- For $f(x) = \frac{1}{2}\|Ax - b\|^2$: $\nabla f = A^\top(Ax - b)$, $\nabla^2 f = A^\top A$.
Use $\eta < 2/\lambda_{\max}(A^\top A)$.

Backtracking Line Search (Armijo Condition)

Choose parameters $\alpha \in (0, 0.5)$ and $\beta \in (0, 1)$. At each step:

1. Start with $\eta = 1$.
2. Check: $f(x_k - \eta \nabla f(x_k)) \leq f(x_k) - \alpha \eta \|\nabla f(x_k)\|^2$
3. If not satisfied: $\eta \leftarrow \beta \eta$ and repeat.
4. Accept the first η that satisfies the condition.

Guarantees sufficient decrease at every step regardless of initial η .

Newton's Method

Root-finding (find $f(x) = 0$):

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

Minimization (find $\nabla f(x) = 0$):

$$x_{k+1} = x_k - [\nabla^2 f(x_k)]^{-1} \nabla f(x_k)$$

Quadratic convergence near the solution (very fast once close).

Projected Gradient Descent (for constraints)

$$x_{k+1} = \Pi_C(x_k - \eta \nabla f(x_k))$$

where $\Pi_C(z) = \arg \min_{x \in C} \|x - z\|$ is the projection onto the feasible set C .

211. LINEAR ALGEBRA ESSENTIALS

Eigenvalues and Eigenvectors

$Av = \lambda v$ (eigenvector v , eigenvalue λ).

Find eigenvalues: solve $\det(A - \lambda I) = 0$.

Positive (Semi-)Definite Matrices

$A \succeq 0$ (PSD) $\Leftrightarrow x^\top A x \geq 0 \forall x \Leftrightarrow$ all eigenvalues $\geq 0 \Leftrightarrow A = B^\top B$ for some B .

$A \succ 0$ (PD) $\Leftrightarrow$ all strict (> 0) $\Leftrightarrow$ unique minimizer for $\min x^\top A x$.

f convex $\Leftrightarrow \nabla^2 f(x) \succeq 0$ for all x .

SVD

$M = U \Sigma V^\top$: U, V orthogonal, Σ diagonal with singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$.

Best rank- k approximation: keep top k singular values (Eckart-Young theorem).

Schur Complement

For block matrix with $C \succ 0$:

$$\begin{pmatrix} A & B \\ B^\top & C \end{pmatrix} \succeq 0 \Leftrightarrow A - B C^{-1} B^\top \succeq 0$$

The matrix $A - BC^{-1}B^\top$ is called the *Schur complement* of C .

-212. TIME-FREQUENCY UNCERTAINTY

Uncertainty principle: $\Delta t \cdot \Delta \omega \geq \frac{1}{2}$

A signal cannot be simultaneously localized in both time and frequency:

- Short time window (spike) $\Rightarrow$ broad frequency spectrum
- Narrow frequency band (sinusoid) $\Rightarrow$ spread over long time

DFT formula:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}, \quad k = 0, 1, \dots, N-1$$

$X[0] = \sum x[n]$ (DC component); $X[N/2]$ = Nyquist (highest frequency).

Problem: DFT gives a global sum — no information about *when* each frequency occurred.

Fix: Short-Time Fourier Transform (STFT) — apply DFT to short overlapping windows.

Example signal $x = [1, -1, 1, -1, 2, 2, 2, 2]$, $N = 8$:

- Samples 1–4: $[1, -1, 1, -1]$ = high-frequency oscillation at $k = 4$ (Nyquist)
- Samples 5–8: $[2, 2, 2, 2]$ = DC (zero frequency, $k = 0$)
- $X[0] = 8$, $X[4] = 4$ — but the DFT cannot tell which came first!

-213. REINFORCEMENT LEARNING

MDP Components: (S, A, P, R, γ)

- S : set of states
- A : set of actions
- $P(s'|s, a)$: transition probability
- $R(s, a)$: reward function
- $\gamma \in [0, 1]$: discount factor

Value Functions

State-value function:

$$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t R_{t+1} \mid s_0 = s \right]$$

Action-value (Q) function:

$$Q^\pi(s, a) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t R_{t+1} \mid s_0 = s, a_0 = a \right]$$

Relation: $V^\pi(s) = \sum_a \pi(a|s) Q^\pi(s, a)$

Bellman Equations

Expectation equation (for a fixed policy π):

$$V^\pi(s) = \sum_a \pi(a|s) \left[R(s, a) + \gamma \sum_{s'} P(s'|s, a) V^\pi(s') \right]$$

Optimality equations (for optimal V^* , Q^*):

$$V^*(s) = \max_a \left[R(s, a) + \gamma \sum_{s'} P(s'|s, a) V^*(s') \right]$$

$$Q^*(s, a) = R(s, a) + \gamma \sum_{s'} P(s'|s, a) \max_{a'} Q^*(s', a')$$

Value Iteration

Initialize V_0 arbitrarily. Repeat until convergence:

$$V_{k+1}(s) = \max_a \left[R(s, a) + \gamma \sum_{s'} P(s'|s, a) V_k(s') \right]$$

Q-Learning Update

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[\underbrace{R + \gamma \max_{a'} Q(s', a')}_{\text{TD target}} - Q(s, a) \right]$$

α : learning rate. Converges if $\sum_t \alpha_t = \infty$ and $\sum_t \alpha_t^2 < \infty$ (Robbins-Monro).

Experience replay: Store (s, a, r, s') transitions in a buffer; sample random minibatches to break temporal correlations.

Policy Evaluation (Matrix Form)

For a fixed π , V^π satisfies the linear system:

$$(I - \gamma P^\pi) V^\pi = R^\pi$$

Solve directly (small MDPs) or iteratively.

-214. SPARSE MATRICES & RL APPLICATIONS

LU Factorization

$A = LU$ (L lower triangular, U upper triangular). Solve $Ax = b$ in two steps:

1. Forward substitution: solve $Ly = b$.
2. Backward substitution: solve $Ux = y$.

Sparse Matrix Storage Formats

A matrix is **sparse** if most entries are zero (real-world: social graphs, circuits).

COO (Coordinate) format — store each nonzero as (i, j, v) :

Memory: $3 \times \text{nnz}$

CSR (Compressed Sparse Row) format:

- `values[]`: nonzero values row by row
- `col_index[]`: column index of each nonzero
- `row_ptr[]`: index in `values[]` where each row starts

Memory: $2 \times \text{nnz} + (n + 1)$ — more compact than COO.

The Fill-in Problem

During LU factorization, row operations can turn zeros into nonzeros — this is **fill-in**. Excessive fill-in destroys sparsity.

Finding the optimal pivot ordering to minimize fill-in is NP-hard.

Solution: Use RL to learn good (approximate) orderings.

RL Formulation for Pivot Ordering

RL Component	Sparse Matrix Equivalent
State	Current nonzero pattern
Action	Choose next pivot row/col
Reward	$-(\# \text{ new fill-in entries created})$
Episode	One full LU factorization

RL-based approach matches or beats all classical heuristics.

Key challenge: Long-term credit assignment — a bad early pivot can cause massive fill-in much later; greedy approaches fail.

Must-Remember Summary

Always True (no conditions):

- Dual function $g(\lambda, \nu)$ is always **concave**
- Conjugate $f^*(y)$ is always **convex**
- Weak duality: $g(\lambda, \nu) \leq p^*$ always
- $f^{**}(x) \leq f(x)$ for any f

Needs Conditions:

- Strong duality $d^* = p^*$: needs convexity + Slater
- KKT sufficient for optimality: needs convexity (+ Slater for necessity)
- Fenchel-Moreau $f^{**} = f$: needs f proper, closed, convex

Quick Formula Recall:

- Comp. slackness: $\lambda_i^* f_i(x^*) = 0$
- Richardson: $\frac{4D(h/2) - D(h)}{3}$
- Q-learning: $Q \leftarrow Q + \alpha[R + \gamma \max_{Q'} Q - Q]$
- CSR memory: $2 \times \text{nnz} + (n + 1)$; fill-in $\Rightarrow$ NP-hard $\Rightarrow$ RL

You know this material. Trust your preparation. Good luck!